



ANTARES ☆

ROBOCUP RESCUE LINE JUNIOR 2026

"Learn • Improve • Compete • Inspire."



1. INTRODUCTION

- First time participating in RoboCup.
- Designed for Rescue Line Junior.
- Capable of following lines, detecting obstacles and autonomously performing actions.
- We learned a lot through trial and error and constant improvement.

2. TEAM MEMBERS



Carlos González
• 16 years old
• Software, programming, claw design, team captain



Ismael Bimonte
• 16 years old
• Hardware, assembly, programming

3. TEAM HISTORY

2025

- Started in robotics with a national SUMO competition

2026

- First participation in RoboCup Rescue Line Junior

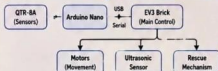


4. HARDWARE OVERVIEW

- LEGO Mindstorms EV3 (main control)
- Arduino Nano
- QTR-8A (line following)
- Ultrasonic sensor (obstacle detection)
- EV3 motors (4-wheel drive)
- 3D printed rescue mechanism with servo motor



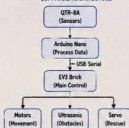
SYSTEM ARCHITECTURE



7. SOFTWARE OVERVIEW </>

- EV3 handles navigation, obstacles and strategies.
- Arduino reads the QTR-8A sensor array.
- Modular software design.
- Easy testing and adjustments.
- Continuous communication EV3 ↔ Arduino.

SOFTWARE ARCHITECTURE



8. INNOVATIVE SOLUTIONS

- Hybrid architecture (EV3 + Arduino Nano).
- Distributed processing for higher efficiency.
- Infrared sensor array for precise line following.
- 3D printed mechanism: lightweight and reliable.
- Simple and robust design for competition.



9. TESTING PROCEDURES

- Line following tests.
- Curves and intersections.
- Obstacle detection tests.
- Rescue mechanism tests.
- Multiple adjustments and optimizations.



10. FUTURE IMPROVEMENTS

- Improve line following algorithm.
- Add new functions and strategies.
- Optimize hardware for more speed and reliability.
- Redesign mechanical parts for better performance.



5. MECHANICAL DESIGN

- LEGO Technic structure
- 4-wheel drive for stability and traction
- 3D printed front claw with servo motor
- Compact, strong and modular design



6. ELECTRONIC DESIGN

- Infrared sensor array (QTR-8A)
- Ultrasonic sensor
- Color sensor (green, red, silver, black, white)
- EV3 motors and servo motor
- Single power source: 9V (EV3 battery)
- EV3 ↔ Arduino: USB Serial communication



11. CONCLUSION

- We put in a lot of effort and achieved most of our goals.
- More organized team and consistency.
- We overcame many challenges and learned a lot as a team.